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Palaeomagnetic studies of a Cretaceous molasse sequence in the central Andes (Coloso Formation, Northern Chile)

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SUMMARY: The Coloso Formation is an early Cretaceous conglomerate-dominated molasse sequence in northern Chile. The sequence was deposited rapidly and records the unroofing of granodioritic and andesitic source rocks. Thin sand units throughout the sequence have been studied palaeomagnetically; the results are complex but can be related to the diagenesis of arid-zone alluvium. Natural remanent magnetization (NRM) may be normal or reversed but is usually a combination of the two. Isolation of magnetic components by vector analysis of thermal and AF demagnetization data yields both normal and reversed components which are approximately antiparallel and yield an average pole position at 70°S, 192°E. This is consistent with late Cretaceous data from northern Chile but not with the established apparent polar wander (APW) path for South America. It may indicate late Cretaceous remagnetization but is considered to be more likely an indication that the present APW path assumes no tectonic rotation and that important relative movements have taken place between northern Chile and Argentina.

The Coloso Formation in Antofagasta Province (Fig. 1) is a conglomerate-dominated alluvial fan succession, 1500 m thick, which passes conformably upwards into evaporites, sandstones and conglomerates which in turn are overlain by Lower Cretaceous marine carbonates. The stratigraphy of the succession has recently been modified (our unpublished data) and the new scheme is employed here (Figs 1 & 2). The conglomerates are generally unstratified and commonly show no internal organization, although crude coarsening or fining-upwards cycles can be recognized locally. Imbrication is developed variably and provides the only palaeocurrent data available. Detailed analysis of imbrication data indicates a source area to the SW. The conglomerates were deposited in proximal and medial fan environments dominated by sheet flood processes. The base of the formation lies with angular unconformity on the andesitic volcanics of the Jurassic La Negra Formation and, near the base, consists of andesitic breccias and conglomerates of local derivation. Progressively upwards through this sequence the proportion of clasts showing propylitic alteration increases and clasts with evidence of copper mineralization are common (Fig. 2). Above this level, clasts of granodioritic composition begin to dominate and this coincides with the occurrence of stratiform copper deposits throughout the upper 150 m of the conglomerate sequence, suggesting a genetic link between framework composition and mineralization. The change in sediment composition seen in the Coloso Formation reflects the reversal of stratigraphy expected during the unroofing of a source area consisting of a parent granodiorite with a volcanic mantle. Many of the deeper level andesites, and especially the granodiorites, show evidence of copper mineralization indicating mineralization of the source rocks. It is to be expected that the erosion of large areas of bedrock mineralization will be reflected at the appropriate level in the unroofing sequence. The thickness of the Coloso Formation conglomerate sequence (1.5 km) is consistent with the unroofing of a stratovolcano/porphyry system (3–6 km) assuming normal sediment dispersal (Sillitoe 1973).

Interbedded with the conglomerates are thin (usually less than 1 m) sand bodies and the lower conglomerate sequence passes into a sand-dominated unit (Fig. 2). These sands frequently show reverse grading and are considered to represent distal portions of the fan bodies which prograded towards the NE. They pass laterally into an upper conglomerate sequence, and together these comprise the Lombriz Formation. The Lombriz Formation is overlain by shallow-marine Neocomian limestones of the Tableado Formation (formerly the Way Formation).

Diagenetic studies of the orebody and associated sediments reveal a number of features characteristic of the shallow burial diagenesis of arid-zone continental sediments (Walker et al. 1978; Turner 1980). These include dissolution and clay replacement of detrital silicate grains and a suite of authigenic minerals including quartz, feldspar, calcite and hematite. In shallow burial diagenesis, reaction between detrital grains and groundwater controls the supply of ions to interstitial pore solutions. Under suitable chemical conditions minerals precipitate as overgrowths, cements or discrete authigenic phases. Detrital framework composition thus exercises an important control on the subsequent diagenesis. In the Coloso Formation sulphide grains and mineralized clasts from the source area have been thoroughly oxidized and replaced by hematite and clearly have been affected by shallow burial diagenesis. The solubility of copper in

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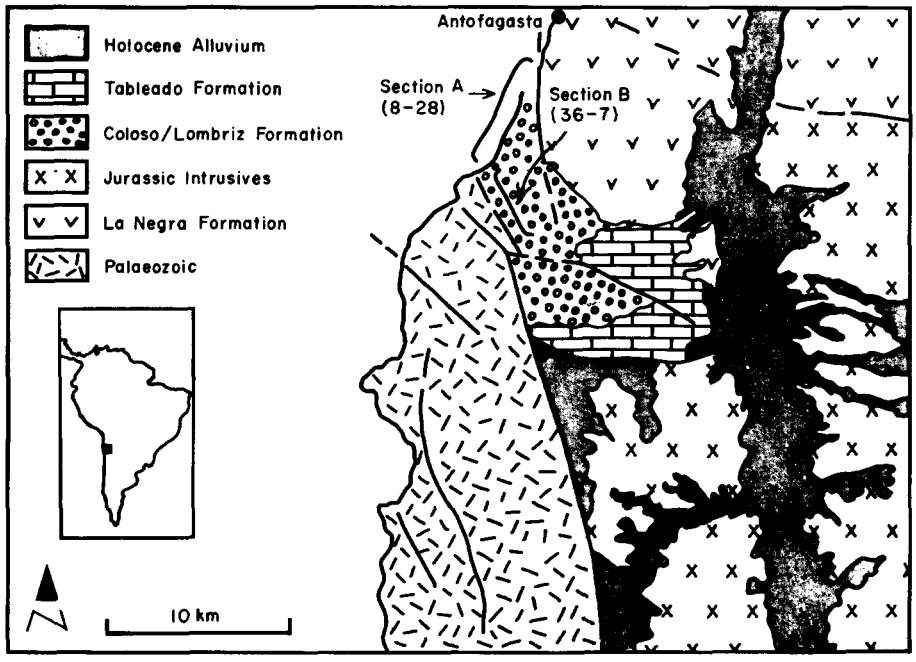


FIG. 1. Geological sketch map of the Antofagasta area, northern Chile showing the location of sections which have been sampled palaeomagnetically.

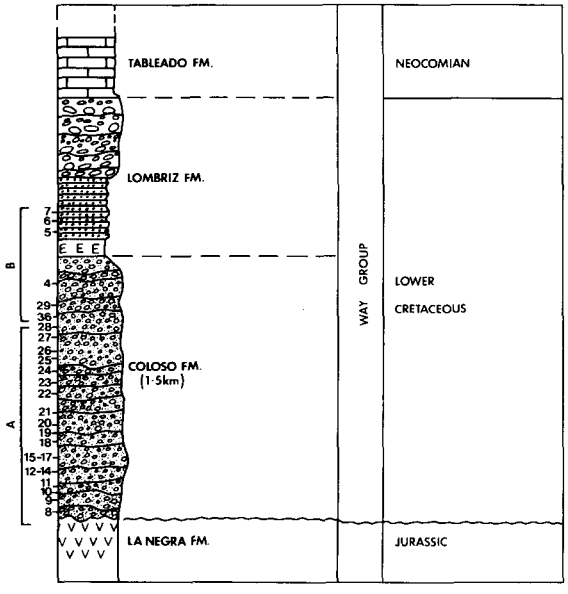


FIG. 2. Generalized profile of the Coloso Formation showing the stratigraphical position of palaeomagnetic samples.

normal terrestrial groundwater is less than 1 ppm near neutral pH but this may increase dramatically if cuprous chloride complexes are formed. The complexion of $CuCl^{2-}$ promotes considerable solubility of copper at intermediate Eh and neutral pH within the stability field of hematite (Rose 1976). Interestingly, cuprous chlorides (atacamite) are present in the mineralized conglomerate orebody in association with the main sulphide phase (covelline).

There have been few palaeomagnetic studies in Chile. Palmer et al. (1980*a*) studied a Late Cretaceous porphyry system in the La Serena area and Jurassic volcanics and sediments near Arica (Palmer et al. 1980*b*). Cretaceous and later palaeomagnetic data from northern Chile are consistent with other South American data, whereas pre-Cretaceous data are not (Valencio et al. 1983). In general, it is not clear how relevant published apparent polar wander (APW) paths for South America are for Andean evolution.

Palaeomagnetic results

Hand samples oriented by sun or magnetic compass were collected from over 40 horizons throughout the Coloso Formation. Two or three specimen cores were

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prepared from each sample and the initial natural remanent magnetization (NRM) was measured with a Digico balanced fluxgate magnetometer. The directions shown in Fig. 3 are predominantly in the NE quadrant with upward dip, although there is a wide degree of scatter.

In order to remove secondary components of magnetization and establish discrete stable components specimens were treated by thermal or alternating field (AF) demagnetization techniques. In most cases, thermal and AF methods have been used for duplicate specimens for the same hand sample so that direct comparison between the blocking temperature spectrum and coercivity spectrum could be made. A variety of types of magnetization were identified in this way and these are categorized as normal (N), reversed (R), intermediate (I) and normal + reversed (N + R). Results are shown in Figs 4–7. Figure 4 shows the results yielded by a typical normally magnetized medium to coarse-grained sandstone from near the base of the Coloso Formation (Fig. 2). The results are shown in three parts: a stereographic projection of the directional changes (A), a normalized intensity decay curve (B) and a Zijderveld orthogonal plot (C) from which the stable component of magnetization for the specimen can best be estimated (Dunlop 1979). Thermal treatment usually reveals that half the NRM is lost between 200°C and 400°C and a marked blocking temperature is often present near 200°C. By 600°C there is usually less than 10% of the original remanence remaining. The blocking temperature spectrum can therefore be described as thermally distributed (Irving & Opdyke 1964). AF treatment also shows marked reduction of the NRM intensity. The median destructive field is about 25 mT and less than 25% of original remanence usually remains after treatment in AF's of 100 mT. Occasional specimens show much greater resistance to both thermal and AF treatments. Such specimens usually show little or no directional change during the demagnetization procedure. Careful inspection of the orthogonal vector plot (Fig. 4c) shows a small degree of curvature in the data points (excluding the first point in the W vs. UP projection), indicating the presence of two closely overlapping components with similar blocking temperature spectra. In such cases, the higher temperature component has been isolated for palaeomagnetic analysis.

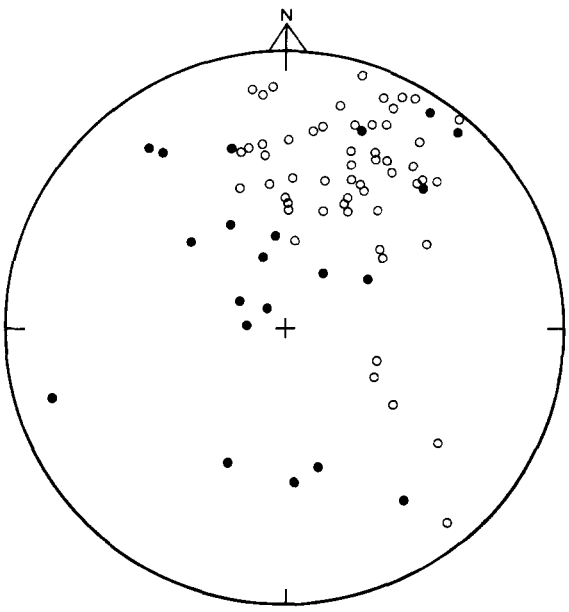


FIG. 3. Stereographic projection showing the widely scattered distribution of initial NRM directions. Open symbols = upward dip; closed symbols = downward dip.

Samples with reversed directions of magnetization show generally similar palaeomagnetic properties. Figure 5 shows two specimens from a sandstone body within the Upper Coloso Formation (see Fig. 2) for location) one of which has been subjected to AF and one to thermal demagnetization. Figure Fig. 5C shows the orthogonal vector plot for the thermal end-points. Again there is a small amount of curvature in the data but the last six data points form a line from which a single non-viscous magnetic component can be estimated.

Samples with more complex magnetizations are common. Generally these take the form of a lower coercivity, lower blocking temperature normal component with a superimposed higher coercivity, higher blocking temperature reversed component. The coercivity 'window' between these two components is usually sufficiently large for the two components to be discriminated readily. The evidence of the presence of two, approximately antiparallel, components of magnetization is clearly seen both on the stereographic projection (directional change) (Fig. 6A) and on the normalized intensity decay curve (Fig. 6 B) where minima are seen in the blocking temperature spectrum (250°C) and coercivity spectrum (30 mT) prior to a relative increase in NRM intensity as the underlying component is isolated. The two components can be roughly estimated from the orthogonal vector plot because of the large window in the coercivity/blocking temperature spectrum. However, it can be seen that in the case of normal magnetization, two closely overlapping sub-components (A and B) can be identified (A = points 2–4; B = points 4–7). In this case the B component, which is better defined, has been used for palaeomagnetic analysis.

In some samples which have antiparallel (N + R) magnetizations there are discrepancies in the behaviour of duplicate specimens which have been

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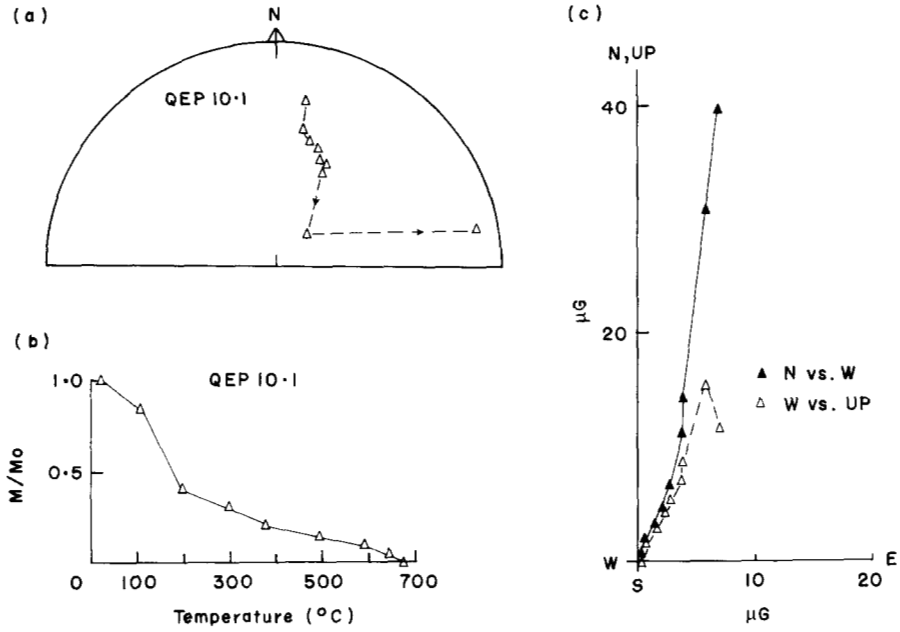


FIG. 4. Thermal demagnetization of a typical normally magnetized sandstone (QEP 10.1). A, Stereographic projection; B, normalized intensity decay; C, orthogonal vector diagram from which the characteristic direction has been measured (see Table 1).

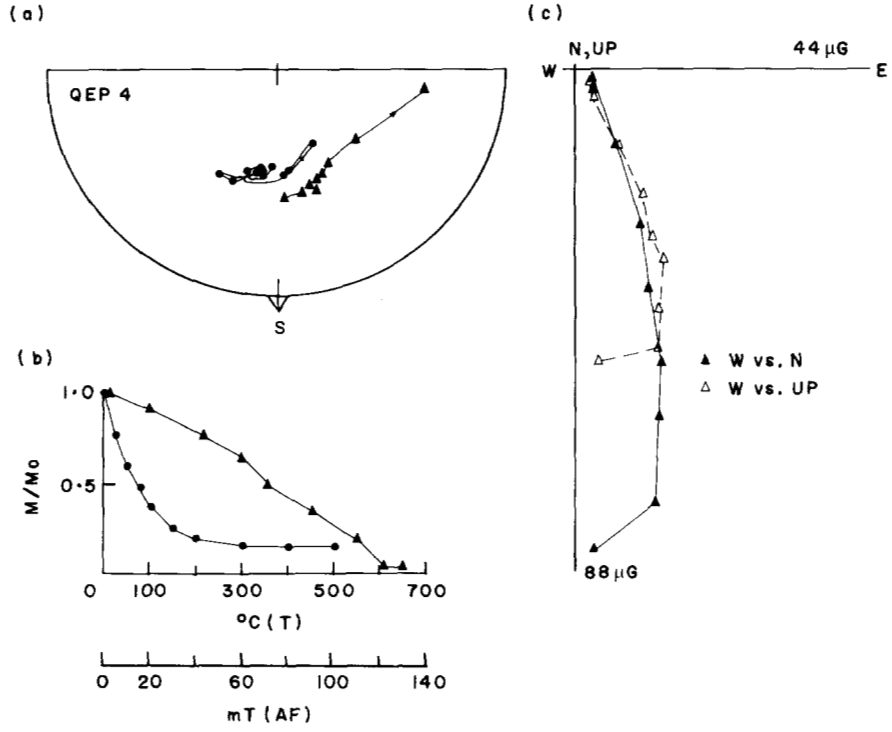


FIG. 5. Thermal (TH) and alternating field (AF) demagnetization results of a typical reversely magnetized sandstone. (QEP4.1, 4.2). Other details as Fig. 4.

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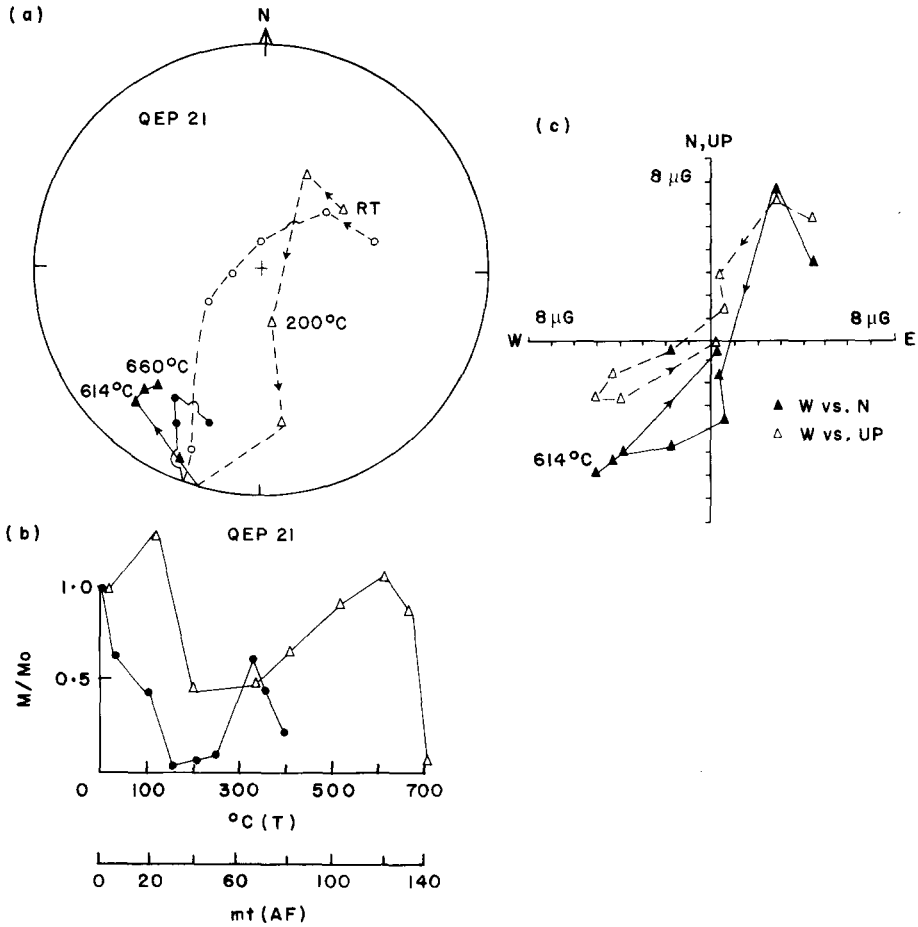


FIG. 6. Thermal (TH) and alternating field (AF) demagnetization results of a sandstone with typical multicomponent magnetization (QEP 21.1, 21.2). The orientation of the isolated components is given in Table 1. Other details as in Fig. 4.

subjected to thermal or AF demagnetization. Figure 7 illustrates a typical case: the thermal data clearly reveal two components which are nearly antiparallel. The AF demagnetization reveals only one normal component of magnetization; in such cases the coercivity is usually greater with median destructive fields in excess of 50 mT. Samples of this type are not considered fundamentally different from those which show N + R components in both AF and thermally demagnetized specimens. They are taken as an indication that the relative proportions of N + R are inhomogeneously distributed; the uncertainties of duplicate sampling may result in selection of specimens in which the subordinate component (R) is poorly developed or absent.

Because of the complexity of the magnetization, conventional 'bulk cleaning' techniques have not been considered suitable for palaeomagnetic analysis. Only components isolated on orthogonal vector plots of partial thermal or AF demagnetization have been used. In cases where a normal or reversed magnetization comprises two or more sub-components only the highest coercivity or highest blocking temperature component has been used for palaeomagnetic analysis. 'Softer' magnetic components have not been used in the palaeomagnetic analysis because they are likely to be secondary in origin. A list of the components used is given in Table 1 and shown in Fig. 8 along with mean pole positions computed by assigning equal weight to all components. Separate poles are listed for both normal and reversed groups within the Coloso Formation. These are very nearly exactly antiparallel, but the normal group is much more precisely defined.

Origin of the NRM

The sandstones of the Coloso Formation have been examined using thin section and polished section

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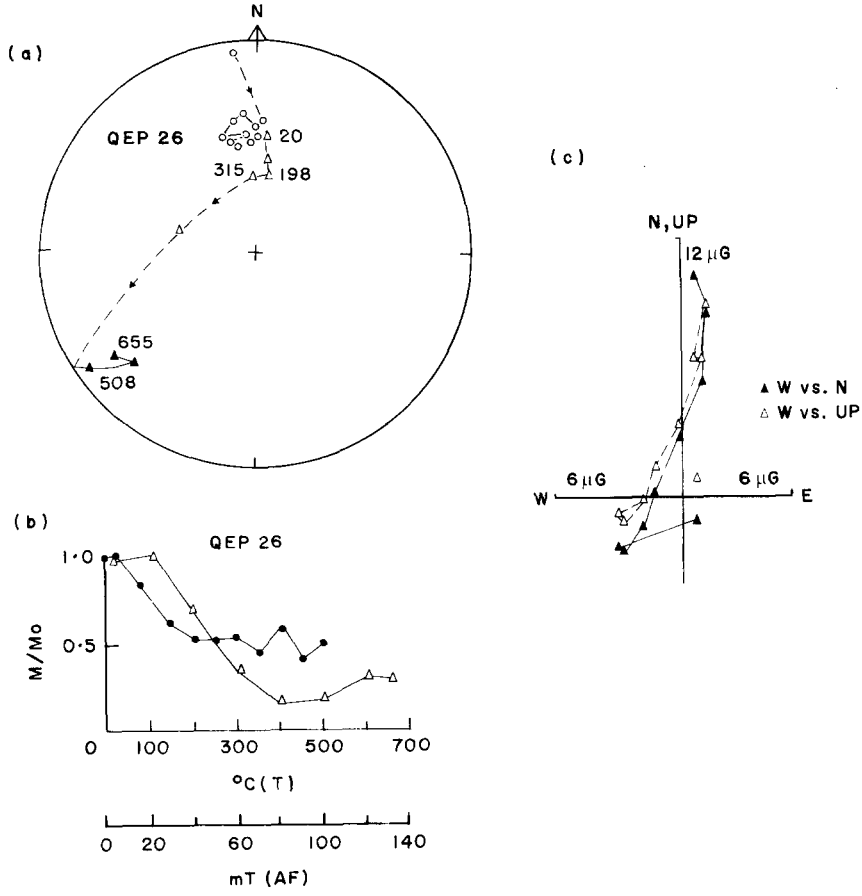


FIG. 7. Thermal and alternating field demagnetization results of a sandstone with multicomponent magnetization (QEP 26.1, 26.2). Note the discrepancy between the thermal and AF directional results. Other details as Fig. 4.

microscopy and SEM. They are lithic arkoses in composition and contain a high proportion of volcanic and intrusive igneous rock fragments. NRM carriers are easily identified because of their abundance and relatively coarse grain size. Polished sections reveal a wide variety of iron–titanium oxides, in various stages of oxidation, which are present both as discrete grains and within detrital rock fragments. The following are the most common in order of abundance (illustrated in Fig. 9).

(A) Martite. These were originally titanomagnetites which have been partially or completely oxidized post-depositionally. They usually show the typical triangular pseudomorphous texture, and frequently contain small irregular relict patches of titanomagnetite or titanomaghemite.

(B) Hemo-ilmenite. This usually includes exolution plates of ilmenite.

(C) Magnetite. Discrete grains of magnetite usually show extensive *in situ* dissolution textures.

(D) Authigenic pigmentary hematite

A variety of other iron–titanium oxides occurs but these are paramagnetics and include ilmenite and various phases of TiO_2. Textural observations indicate that oxidation took place *in situ* and prior to the main phase of sulphide mineralization in the Upper Coloso Formation.

The NRM of the Coloso sandstones can be interpreted as being carried by both hematite and magnetite. The large difference in coercivity/blocking temperature spectra seen in specimens with multicomponent magnetization suggests that both hematite and magnetite carry NRM components. However lower coercivity/lower blocking temperature components (which are usually N-polarity) are probably carried mainly by magnetite. This magnetization must

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TABLE 1: *Palaeomagnetic direction isolated by thermal/alternating field demagnetization from the Coloso Formation samples*

Normal components				*Reversed components*			
Specimen	*D*	I	*Treatment*	*Specimen*	*D*	I	*Treatment*
QEP 5.2*	33	−34	AF	QEP 4.1	163	36	TH
QEP 7.2*	10	−21	AF	QEP 4.2	191	42	AF
QEP 7.3*	0	−35	TH	QEP 11.1	244	12	AF
QEP 8.2	42	−19	AF	QEP 14.1	228	55	AF
QEP 10.1	9	−23	TH	QEP 21.1	222	18	TH
QEP 14.1	15	−53	AF	QEP 21.2	224	18	TH
QEP 17.1	34	−45	AF	QEP 21.3	178	23	AF
QEP 21.2	24	−54	TH	QEP 26.1	243	23	AF
QEP 23.1	15	−39	TH	QEP 26.2	246	25	TH
QEP 23.2	40	−26	TH				
QEP 24.2	42	−44	TH				
QEP 25.1	25	−36	TH				
QEP 26.2	18	−38	TH				
QEP 26.2A	20	−38	AF				
QEP 27.1	10	−30	TH				
QEP 27.2A	16	−24	AF				
QEP 28.1	0	−35	TH				
Mean $\bar{D} = 21$; $\bar{I} = -36$; $n = 17$; $R = 16.435$; $K = 28.3$; CSD = 15.2; $\alpha_{95} = 6.8$, pole = lat.: 70°S, long.: 192°E.				Mean $\bar{D} = 217$; $\bar{I} = 31$; $n = 9$; $R = 7.948$; $K = 7.6$; CSD = 29.4; $\alpha_{95} = 20.0$; pole = lat.: 55°S, long.: 195°E.			

* From the overlying Lombriz Formation.

R = Vector magnitude; CSD = Circular standard deviation; α_{95} = 95% cone of confidence; K = Precision parameter.

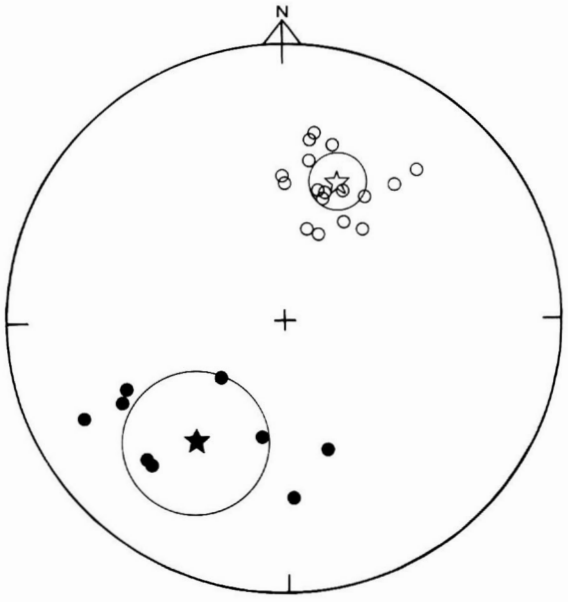


FIG. 8. Stereographic projection showing the overall distribution of components isolated by thermal and AF demagnetization. The components are listed in Table 1. The mean and α_{95} of both normal and reversed clusters is also shown.

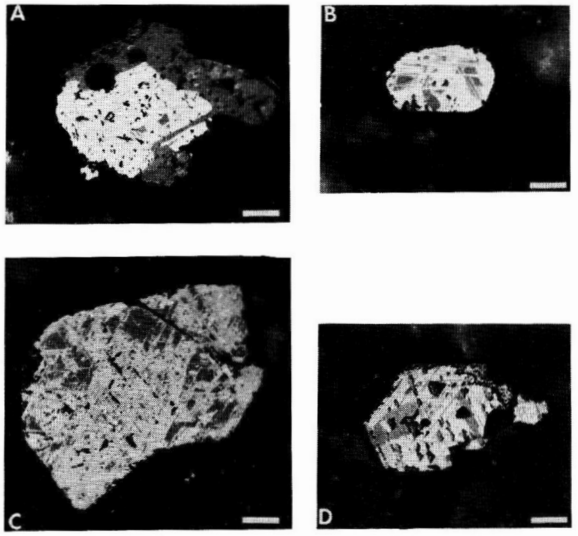


FIG. 9. Reflected light photomicrographs of iron-titanium oxides involved in multicomponent magnetization in the Coloso Formation. A, Martite, with small inclusions of titanomagnetite rimmed by large areas of ilmenite; B, martite showing discrete areas of relict titanomagnetite; C, large martite with diffuse patches of relict titanomagnetite; D, martite with large sheets and irregular patches of titanomagnetite. Bar scale = 80 μm.

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originally have been associated with deposition or early diagenesis. As diagenesis progressed detrital titanomagnetites were oxidized post-depositionally and magnetite was converted to maghemite or hematite. Many grains (Fig. 9) show compound mineralogy and it is these which can be expected to account for the coexisting normal and reversed directions of magnetization. The fact that the higher blocking temperature components (carried at least partly by hematite) are exclusively reversed polarity suggests that this diagenetic magnetization may have been generated by a discrete phase of oxidation associated with a period of non-deposition which followed deposition of the Upper Coloso Formation (sandstone units). Secondary (i.e. post-diagenetic) magnetizations subsequently may have been acquired, particularly by coarse-grained (multidomain) magnetite which survived diagenetic oxidation. These magnetizations are predominantly normal and could represent viscous components or partial thermal-remanent magnetizations associated with burial and uplift or Andean igneous activity.

Discussion

The pole position defined by the group of components with normal directions has the lowest α_{95} (Table 1) and is considered to provide the best estimate for the Coloso Formation. This pole (70°S, 192°E) is similar to other Cretaceous data from northern Chile although these are from the late Cretaceous. Available data for South America as a whole imply rather rapid apparent polar wandering and substantial differences between early and late Cretaceous pole positions (Valencio *et al.* 1983). When compared with the apparent polar wander path for South America the Coloso pole lies nearest to the hairpin defined by poles of Permian to mid-Jurassic age. Although the age of the Coloso Formation is not known precisely it is clear that the bulk of the formation must be post late Jurassic because it is conformably overlain by Neocomian limestones. There remain, therefore, two alternatives: (a) the NRM of the Coloso Formation was remagnetized in late Cretaceous time, perhaps in association with porphyry intrusion; or (b) there are minor structural differences between the western flank of the Andes and the stable area of Argentina (the principal source of South American data).

Support for the second alternative comes from the fact that there are discrepancies between palaeomagnetic directions from the mid-Jurassic in northern Chile (Palmer *et al.* 1980*b*) and elsewhere in South America. These are interpreted as the result of rotational or strike-slip movements prior to the opening of the South Atlantic. They could equally well be associated with accretionary tectonics on the western flank of the Andes. On balance, it seems likely that the published APW path for South America is not applicable to northern Chile.

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